A Hardy space approximation supporting zero-free half-planes for the ζ-function

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Abstract

Bagchi's Hilbert-space reformulation of the Báez-Duarte criterion [2] for the Riemann Hypothesis (RH) [5] states that RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of $\{G_k : k \geq 2\}$, where $G_k(s) = (k^{-s} - k^{-1})\zeta(s)/s$, in the Hardy space $H^2(\mathbb{C}_{1/2})$; here $\mathbb{C}_\alpha$ denotes the half-plane $\mathrm{Re}(s) > \alpha$. We first prove that the series $\sum_{k=2}^{\infty} \mu(k)G_k$, with μ the Möbius function, converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$, which in particular gives a new proof that ζ is zero-free in $\mathbb{C}_1$. We also show that if E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$ for some $\alpha \geq 1/2$, then ζ is zero-free in $\mathbb{C}_\alpha$. Motivated by this, we carry out a numerical study of the approximation error

$$s_n = \left\| \sum_{k=2}^{n} \mu(k)G_k - E \right\|_\alpha^2, \qquad 1/2 \leq \alpha \leq 1,$$

where $\|\cdot\|_\alpha$ is the norm of $H^2(\mathbb{C}_\alpha)$. The data suggest convergence for α near 1 and persistent oscillation near $\alpha = 1/2$, consistent with the known divergence on the critical line.

Keywords: Riemann hypothesis, zero-free half-planes, Báez-Duarte criterion, Hardy spaces, numerical approximation.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$. Its implications for the distribution of prime numbers make it one of the central open problems in mathematics.

A functional-analytic approach to RH was initiated by Nyman and Beurling, who showed that RH is equivalent to the density of a specific family of functions in $L^2(0,1)$ [6, 11]. Báez-Duarte

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later refined this by replacing the original uncountable family with a countable sequence, simplifying the problem while preserving its essential structure [2]. Balazard and Saias [4] extended this approach to a weighted ℓ^2-sequence space by posing the question of whether the orthogonal projection coefficients $c_{n,k}$ of $\mathbf{1}$ onto $\operatorname{span}\{r_k : 2 \le k \le n\}$ satisfy

$$\lim_{n\to\infty} c_{n,k} = -\frac{\mu(k)}{k}, \quad \forall\, k \ge 2,$$

where μ is the Möbius function and $r_k(j) = j \bmod k$ denotes the remainder upon division of j by k. This condition was shown by Weingartner in [13] to be equivalent to the RH, which identifies $-\mu(k)/k$ as the natural, and in a precise sense optimal, choice of coefficients (equivalently $\mu(k)$ after the normalization used below).

A reformulation within the Hardy space $H^2(\mathbb{C}_{1/2})$ was established by Bagchi [5], where $\mathbb{C}_\alpha = \{s \in \mathbb{C} : \operatorname{Re}(s) > \alpha\}$ for $\alpha \in \mathbb{R}$: RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of

$$G_k(s) = \left(k^{-s} - k^{-1}\right)\frac{\zeta(s)}{s}, \quad k \ge 2,$$

in $H^2(\mathbb{C}_{1/2})$. Building on these ideas, Noor introduced an equivalent reformulation in the Hardy–Hilbert space $H^2(\mathbb{D})$ of the unit disk [10]: RH holds if and only if the constant function $\mathbf{1}$ belongs to the closed linear span of

$$h_k(z) = \frac{1}{1-z}\log\left(\frac{1 + z + \cdots + z^{k-1}}{k}\right), \quad k \ge 2.$$

In the same work, Noor established that the series $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I - S)h_k$ converges to $1 - z$ in $H^2(\mathbb{D})$, where S is the shift operator. For an overview of the various equivalent reformulations of the Nyman-Beurling and Báez-Duarte approaches that appear in the literature, see [7, p. 5-6].

The main idea of this paper is to generalize this result in an attempt to approximate RH using the reformulation of Bagchi in $H^2(\mathbb{C}_{1/2})$. To achieve this, we employ the Hilbert space of Dirichlet series $\mathcal{H}^2$, which presents a natural connection with the Hardy space of the disk. Within this framework, we construct a bounded operator $\mathcal{M}^\tau$ that establishes a connection between $\mathcal{H}^2$ and the Hardy spaces $H^2(\mathbb{C}_\alpha)$, leading to our main result (Theorem 3.1): the series $\sum_{k=2}^{\infty} \mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$. Furthermore, Proposition 3.1 establishes that for any $\alpha \ge 1/2$,

$$E \in \overline{\operatorname{span}}\{G_k : k \ge 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \ne 0, \quad \forall\, s \in \mathbb{C}_\alpha.$$

An important question that arises is whether the convergence of the series extends to $1/2 \le \alpha \le 1$. Báez-Duarte [3] showed that the $L^2(0,1)$-analog of the sequence s_n defined below diverges, which implies the divergence of s_n itself in $H^2(\mathbb{C}_{1/2})$. However, this does not discount the possibility of some subsequence of s_n converging to 0. In fact, the numerical evidence we present below strongly suggests that such subsequences exist. Studying the convergence in the range $1/2 < \alpha \le 1$ is therefore essential, as it could provide information about zero-free regions of ζ within the critical strip, a problem that remains open and is widely considered to be as difficult as the RH itself.

The numerical analysis conducted in this work investigates the behavior of the partial sums

$$s_n = \left\| \sum_{k=2}^{n} \mu(k)G_k - E \right\|^2_{H^2(\mathbb{C}_\alpha)}$$

for $0.5 \leq \alpha \leq 1.0$: the sequence exhibits a clear decreasing trend for α close to 1 and persistent oscillations for α close to 0.5, the latter consistent with the divergence result of Báez-Duarte. Section 2 introduces the necessary function spaces and key results. Section 3 develops the main theoretical framework. Section 4 presents the numerical experiments and their implications.

2 Preliminaries

We collect the function spaces and key results needed throughout the paper. The three Hilbert spaces $H^2(\mathbb{D})$, $\mathcal{H}^2$, and $H^2(\mathbb{C}_{1/2})$ are introduced here as the analytic framework within which the completeness conditions equivalent to the Riemann Hypothesis are formulated and studied.

2.1 Hardy–Hilbert spaces and Dirichlet series

Let $H^2(\mathbb{D})$ be the Hardy–Hilbert space of the unit disk, defined as

$$H^2(\mathbb{D}) = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \|f\|_{H^2}^2 = \sum_{n=0}^{\infty} |a_n|^2 < \infty \right\}.$$

A recent reformulation of the RH, introduced in [10], is stated as follows. For each $k \geq 2$, define

$$h_k(z) = \frac{1}{1-z} \log\left(\frac{1 + z + \cdots + z^{k-1}}{k} \right).$$

Theorem 2.1 (Theorem 6, [10]). *The Riemann Hypothesis holds if and only if the constant function* $\mathbf{1}$ *belongs to the closed linear span of* $\{h_k : k \geq 2\}$ *in* $H^2(\mathbb{D})$.

Let $S = M_z$ denote the shift operator on $H^2(\mathbb{D})$, corresponding to multiplication by z. An additional result from [10] plays a central role in this paper.

Lemma 2.1 (Lemma 11, [10]). *The series* $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I - S)h_k$ *converges to* $1 - z$ *in* $H^2(\mathbb{D})$, *where* μ *is the Möbius function.*

Let $\mathcal{H}^2$ denote the Hilbert space of Dirichlet series [8], defined as

$$\mathcal{H}^2 = \left\{ f(s) = \sum_{n=1}^{\infty} a_n n^{-s} : \|f\|_{\mathcal{H}^2}^2 = \sum_{n=1}^{\infty} |a_n|^2 < \infty \right\}.$$

Setting $H_0^2(\mathbb{D}) = H^2(\mathbb{D}) \ominus \mathbb{C}$, there exists an isometric isomorphism [8]

$$\psi : H_0^2(\mathbb{D}) \longrightarrow \mathcal{H}^2, \qquad z^k \longmapsto k^{-s}.$$

2.2 The Hardy spaces $H^2(\mathbb{C}_\alpha)$

For $\alpha \in \mathbb{R}$, let $\mathbb{C}_\alpha = \{s = \sigma + it : \sigma > \alpha\}$ and let $H^2(\mathbb{C}_\alpha)$ denote the Hardy space of analytic functions F on $\mathbb{C}_\alpha$ satisfying

$$\|F\|_{H^2(\mathbb{C}_\alpha)}^2 := \sup_{x > \alpha} \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(x + it)|^2 \, dt < \infty.$$

Any such F admits a non-tangential boundary value F^* almost everywhere on $\{\mathrm{Re}(s) = \alpha\}$, with

$$\|F\|^2_{H^2(\mathbb{C}_\alpha)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F^*(\alpha + it)|^2\, dt.$$

Moreover, $H^2(\mathbb{C}_\alpha)$ is a Hilbert space with reproducing kernel

$$k_w(s) = \frac{1}{s + \overline{w} - 2\alpha}, \quad w \in \mathbb{C}_\alpha;$$

in particular, convergence in norm implies pointwise convergence in $\mathbb{C}_\alpha$. For brevity we write $\|\cdot\|_\alpha := \|\cdot\|_{H^2(\mathbb{C}_\alpha)}$.

Define the functions G_k and E in $H^2(\mathbb{C}_{1/2})$, for $k \geq 2$, as

$$G_k(s) = (k^{-s} - k^{-1})\frac{\zeta(s)}{s} \quad \text{and} \quad E(s) = \frac{1}{s}.$$

The Riemann Hypothesis reformulation for $H^2(\mathbb{C}_{1/2})$ can be stated as follows.

Theorem 2.2 (Theorem 2, [5]). *RH is true if and only if E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_{1/2})$.*

3 An $H^2(\mathbb{C}_\alpha)$ approximation approach

The main goal of this section is to show that E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$. For $\tau > 1/2$ and $k \geq 2$, define

$$g_{k,\tau}(z) = k^{\tau-1}\sum_{n=1}^{\infty}\frac{z^n}{n^\tau} - \sum_{n=1}^{\infty}\frac{z^{nk}}{n^\tau}.$$

It is clear that $g_{k,\tau} \in H^2_0$. Furthermore, we observe that

$$\begin{aligned}(\psi g_{k,\tau})(s) &= k^{\tau-1}\sum_{n=1}^{\infty}\frac{n^{-s}}{n^\tau} - k^{-s}\sum_{n=1}^{\infty}\frac{n^{-s}}{n^\tau}\\ &= (k^{\tau-1} - k^{-s})\zeta(s+\tau)\\ &= k^{\tau-1}f_{k,\tau}(s) \quad \text{for } \mathrm{Re}(s) > 1/2,\end{aligned}$$

where $f_{k,\tau}(s) = (1 - k^{1-s-\tau})\zeta(s+\tau)$.

3.1 A bounded operator from $\mathcal{H}^2$ to $H^2(\mathbb{C}_\alpha)$

The operator $\mathcal{M}_{1/s} : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2})$, defined by $\mathcal{M}_{1/s} : f(s) \mapsto f(s)/s$, is bounded according to [12]. For $\tau \geq 1/2$, we extend this construction by defining $\mathcal{M}^\tau : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2+\tau})$ via $\mathcal{M}^\tau : f \longmapsto f(s-\tau)/s$. To see that $\mathcal{M}^\tau$ is well defined and bounded, note that for $\sigma > \tau + 1/2$,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{|f(\sigma-\tau+it)|^2}{|\sigma+it|^2}\,dt \leq \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{|f(\sigma-\tau+it)|^2}{|\sigma-\tau+it|^2}\,dt \leq \|\mathcal{M}_{1/s}f\|^2_{H^2(\mathbb{C}_{1/2})},$$

and taking the supremum over $\sigma > \tau + 1/2$ yields

$$\|\mathcal{M}^\tau f\|_{H^2(\mathbb{C}_{1/2+\tau})} \le \|\mathcal{M}_{1/s}\| \, \|f\|_{\mathcal{H}^2}, \qquad \forall f \in \mathcal{H}^2.$$

The relevance of $\mathcal{M}^\tau$ to our approach lies in how it acts on the functions 1 and $f_{k,\tau}$. A direct computation gives

$$(\mathcal{M}^\tau 1)(s) = \frac{1}{s} = E(s),$$

and

$$(\mathcal{M}^\tau f_{k,\tau})(s) = \frac{f_{k,\tau}(s-\tau)}{s} = (1 - k^{1-s})\frac{\zeta(s)}{s} = -kG_k(s).$$

3.2 Zero free region of ζ in $\mathbb{C}_\alpha$

We now establish one of the main results of the paper. The key step is the following generalization of Lemma 2.1 to the family $\{g_{k,\tau}\}$.

Lemma 3.1. *The series $\sum_{k=2}^\infty \frac{\mu(k)}{k^\tau} g_{k,\tau}$ converges to $-z$ in H_0^2, for each $\tau > 1/2$.*

Proof. Since $g_{1,\tau} \equiv 0$, the $k = 1$ term may be included, giving

$$\begin{aligned}\sum_{k=2}^n \frac{\mu(k)}{k^\tau} g_{k,\tau}(z) &= \sum_{k=1}^n \frac{\mu(k)}{k^\tau} \left\{ k^{\tau-1} \sum_{j=1}^\infty \frac{z^j}{j^\tau} - \sum_{j=1}^\infty \frac{z^{jk}}{j^\tau} \right\} \\ &= \sum_{k=1}^n \frac{\mu(k)}{k} \sum_{j=1}^\infty \frac{z^j}{j^\tau} - \sum_{k=1}^n \frac{\mu(k)}{k^\tau} \sum_{j=1}^\infty \frac{z^{jk}}{j^\tau}.\end{aligned}$$

Since $\sum_{k=1}^\infty \frac{\mu(k)}{k} = 0$, the first term vanishes in the limit:

$$\left\| \sum_{k=1}^n \frac{\mu(k)}{k} \sum_{j=1}^\infty \frac{z^j}{j^\tau} \right\|_{H_0^2} = \left| \sum_{k=1}^n \frac{\mu(k)}{k} \right| \left\| \sum_{j=1}^\infty \frac{z^j}{j^\tau} \right\|_{H_0^2} \longrightarrow 0.$$

It therefore suffices to show that

$$\left\| -\sum_{k=1}^n \frac{\mu(k)}{k^\tau} \sum_{j=1}^\infty \frac{z^{jk}}{j^\tau} + z \right\|_{H_0^2} \longrightarrow 0. \tag{3.1}$$

A rearrangement of the double sum gives

$$\begin{aligned}\sum_{k=1}^n \frac{\mu(k)}{k^\tau} \sum_{j=1}^\infty \frac{z^{jk}}{j^\tau} &= \sum_{j=1}^n \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) + \sum_{j=n+1}^\infty \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) \\ &= \sum_{j=1}^n \frac{z^j}{j^\tau} \left[\frac{1}{j} \right] + \sum_{j=n+1}^\infty \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right), \end{aligned} \tag{3.2}$$

where the last equality uses the identity $\sum_{d|j} \mu(d) = \left[\frac{1}{j}\right]$. From (3.2),

$$\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} = z + \phi_n(z),$$

where

$$\phi_n(z) = \sum_{j=n+1}^{\infty} \left(\frac{1}{j^\tau} \sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) z^j.$$

It remains to show that $\|\phi_n\|_{H_0^2} \to 0$ as $n \to \infty$. Letting $d(n)$ denote the number of divisors of n, we estimate

$$\|\phi_n\|_{H_0^2}^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left| \sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right|^2 \le \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left(\sum_{d|j} 1 \right)^2 = \sum_{j=n+1}^{\infty} \frac{d(j)^2}{j^{2\tau}}.$$

Since $d(n) = o(n^\epsilon)$ for every $\epsilon > 0$, choosing $\epsilon > 0$ such that $\tau > 1/2 + \epsilon$ yields

$$\|\phi_n\|_{H_0^2}^2 \lesssim \sum_{j=n+1}^{\infty} \frac{j^{2\epsilon}}{j^{2\tau}} = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau - 2\epsilon}} \longrightarrow 0,$$

since $2\tau - 2\epsilon > 1$. □

Theorem 3.1. *The series $\sum_{k=2}^{\infty} \mu(k) G_k$ converges to E in $H^2(\mathbb{C}_{1/2+\tau})$, for every $\tau > 1/2$.*

Proof. By Lemma 3.1 and the isomorphism ψ,

$$\sum_{k=2}^{n} \frac{\mu(k)}{k} f_{k,\tau} \longrightarrow -1 \quad \text{in } \mathcal{H}^2.$$

Applying $\mathcal{M}^\tau$ and using $(\mathcal{M}^\tau f_{k,\tau})(s) = -k G_k(s)$, we conclude

$$\sum_{k=2}^{n} \mu(k) G_k \longrightarrow E \quad \text{in } H^2(\mathbb{C}_{1/2+\tau}). \tag{3.3}$$

□

Corollary 3.1. *E belongs to the closed linear span of $\{G_k : k \ge 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$.*

The membership of E in the closed linear span of $\{G_k : k \ge 2\}$ also has direct consequences for the location of the zeros of ζ.

Proposition 3.1. *For any $\alpha \ge 1/2$,*

$$E \in \overline{\text{span}}\{G_k : k \ge 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \ne 0, \quad \forall s \in \mathbb{C}_\alpha.$$

Proof. Let ρ be a zero of ζ in $\mathbb{C}_\alpha$. Then $G_k(\rho) = 0$ for every $k \ge 2$. Since $H^2(\mathbb{C}_\alpha)$ is a reproducing kernel Hilbert space, convergence in norm implies pointwise convergence, so $E(\rho) = 1/\rho = 0$, which is impossible. □

3.3 A lower bound from the zeros of ζ

Proposition 3.1 is qualitative: it turns convergence into a zero-free half-plane. The reproducing kernel of $H^2(\mathbb{C}_\alpha)$ yields a quantitative companion in the opposite direction, in which a single zero of ζ inside $\mathbb{C}_\alpha$ produces an explicit, n-independent floor for the approximation error. For $n \geq 2$ and $\alpha \in \mathbb{R}$, define the *optimal approximation error*

$$d_n(\alpha)^2 := \inf_{c_2,\dots,c_n \in \mathbb{C}} \Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2,$$

the best approximation of E by the first $n-1$ generators. Taking $c_k = \mu(k)$ shows $d_n(\alpha)^2 \leq \|S_n - E\|_\alpha^2$, so any lower bound for $d_n(\alpha)^2$ is a fortiori one for the Möbius error.

Theorem 3.2. *Let $\alpha \in \mathbb{R}$ and suppose ζ has a zero ρ with $\mathrm{Re}(\rho) > \alpha$. Then for every $n \geq 2$ and all coefficients $c_2, \dots, c_n \in \mathbb{C}$,*

$$\Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2 \;\geq\; \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2} \;=\; \frac{2\,(\mathrm{Re}\,\rho - \alpha)}{|\rho|^2} \;>\; 0.$$

More generally, if $\rho_1, \dots, \rho_m$ are zeros of ζ in $\mathbb{C}_\alpha$, then for every $n \geq 2$,

$$d_n(\alpha)^2 \;\geq\; v^* K^{-1} v, \qquad K = \Big(\tfrac{1}{\rho_i + \overline{\rho_j} - 2\alpha} \Big)_{i,j=1}^{m}, \qquad v = \big(1/\rho_i\big)_{i=1}^{m}.$$

In particular $d_n(\alpha)^2$ is bounded below by a positive constant independent of n, so $d_n(\alpha)^2 \not\to 0$.

Proof. Since $\mathrm{Re}(\rho) > \alpha$ we have $\rho \in \mathbb{C}_\alpha$, and $\zeta(\rho) = 0$ gives $G_k(\rho) = (k^{-\rho} - k^{-1})\zeta(\rho)/\rho = 0$ for every k. Hence $F := E - \sum_{k=2}^{n} c_k G_k$ satisfies $F(\rho) = E(\rho) = 1/\rho$. By the reproducing property $F(\rho) = \langle F, k_\rho \rangle_\alpha$, so the Cauchy–Schwarz inequality gives

$$\|F\|_\alpha^2 \;\geq\; \frac{|\langle F, k_\rho \rangle_\alpha|^2}{\|k_\rho\|_\alpha^2} = \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2}.$$

With $\|k_\rho\|_\alpha^2 = k_\rho(\rho) = (2\,\mathrm{Re}\,\rho - 2\alpha)^{-1}$ and $|E(\rho)|^2 = |\rho|^{-2}$, this is the first inequality.

For the general case, let $N = \{ g \in H^2(\mathbb{C}_\alpha) : g(\rho_i) = 0,\ 1 \leq i \leq m \}$, whose orthogonal complement is $V = \mathrm{span}\{k_{\rho_1}, \dots, k_{\rho_m}\}$. Every combination $\sum_k c_k G_k$ lies in N, since $G_k(\rho_i) = 0$; thus, with P the orthogonal projection onto V,

$$\Big\| E - \sum_k c_k G_k \Big\|_\alpha \;\geq\; \Big\| P\big(E - \textstyle\sum_k c_k G_k\big) \Big\|_\alpha = \|PE\|_\alpha.$$

A standard computation with the kernels gives $\|PE\|_\alpha^2 = v^* K^{-1} v$, where $K_{ij} = \langle k_{\rho_j}, k_{\rho_i} \rangle_\alpha = k_{\rho_j}(\rho_i) = (\rho_i + \overline{\rho_j} - 2\alpha)^{-1}$ and $v_i = \langle E, k_{\rho_i} \rangle_\alpha = \overline{E(\rho_i)}$. Taking the infimum over c yields the bound on $d_n(\alpha)^2$; the right-hand side is a fixed positive constant, so $d_n(\alpha)^2 \not\to 0$. □

Lemma 3.2. *Let $\frac{1}{2} \leq \alpha \leq \beta$. Then $H^2(\mathbb{C}_\alpha) \subseteq H^2(\mathbb{C}_\beta)$ by restriction, with $\|F\|_\beta \leq \|F\|_\alpha$; consequently $d_n(\beta) \leq d_n(\alpha)$ for every $n \geq 2$.*

Proof. As $\mathbb{C}_\beta \subseteq \mathbb{C}_\alpha$, the supremum defining $\|\cdot\|_\beta$ runs over a subfamily of the vertical lines occurring in $\|\cdot\|_\alpha$. Apply this to $E - \sum_{k=2}^{n} c_k G_k$ and take the infimum over c. □

Corollary 3.2. *Let* $\Theta = \sup\{\operatorname{Re}\rho : \zeta(\rho) = 0,\ \rho \text{ non-trivial}\}$ *and*

$$\alpha_c = \inf\big\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\big\}.$$

Then $\frac{1}{2} \leq \Theta \leq \alpha_c \leq 1$, *and* $\alpha_c = \frac{1}{2}$ *if and only if the Riemann Hypothesis holds.*

Proof. By Lemma 3.2 the set $\{\alpha \geq \frac{1}{2} : d_n(\alpha)^2 \to 0\}$ is upward closed, so α_c is an abscissa. For $\alpha < \Theta$ pick a zero ρ with $\operatorname{Re}\rho > \alpha$; Theorem 3.2 gives $d_n(\alpha)^2 \not\to 0$, whence $\Theta \leq \alpha_c$. Corollary 3.1 gives $\alpha_c \leq 1$, and $\Theta \geq \frac{1}{2}$ is classical. If $\alpha_c = \frac{1}{2}$ then $\Theta = \frac{1}{2}$. Conversely, under RH Theorem 2.2 places E in the closed span of $\{G_k\}_{k\geq 2}$ in $H^2(\mathbb{C}_{1/2})$; as $d_n(1/2)$ is non-increasing in n, any finite combination approximating E forces $d_n(1/2)^2 \to 0$, and Lemma 3.2 propagates this to every $\alpha \geq \frac{1}{2}$. □

4 Numerical experiments

Theorem 3.1 establishes that $\sum \mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$, while Proposition 3.1 shows that the membership of E in the closed linear span of $\{G_k : k \geq 2\}$ forces ζ to be zero-free on $\mathbb{C}_\alpha$. The convergence is settled for $\alpha > 1$; the range $1/2 \leq \alpha \leq 1$ (precisely the portion of the critical strip where the behaviour of ζ is least understood) lies beyond the reach of our analytical methods. The purpose of this section is to probe this range numerically.

Two questions guide the experiments. The first is whether the convergence of the partial sums $S_n = \sum_{k=2}^{n} \mu(k)G_k$ to E persists in the range $1/2 < \alpha \leq 1$. To this end we monitor the approximation error $s_n = \|S_n - E\|_\alpha^2$: by Proposition 3.1, numerical evidence that $s_n \to 0$ for a given α is evidence in favour of the non-vanishing of ζ on $\mathbb{C}_\alpha$, while at the opposite extreme the $L^2(0,1)$-divergence established by Báez-Duarte [3] propagates to s_n at $\alpha = 1/2$, so that genuine convergence cannot be expected on the critical line itself. The second question is whether the obstruction at $\alpha = 1/2$ is intrinsic to the subspaces $\operatorname{span}\{G_2, \ldots, G_n\}$, or is instead an artefact of the fixed choice of coefficients $\mu(k)$; here the optimal error $d_n(\alpha)$ of Subsection 3.3 separates the two possibilities. Subsections 4.2 and 4.3 address the first question, Subsection 4.4 the second, and Subsection 4.5 summarises the findings. We begin by describing the computational methodology common to all experiments.

4.1 Numerical method

Two quantities are computed throughout. The first is the Möbius error $s_n = \|S_n - E\|_\alpha^2$, obtained by numerically evaluating the boundary integral that defines the $H^2(\mathbb{C}_\alpha)$ norm of $S_n - E$. This integral was computed by adaptive Gauss–Kronrod quadrature (the QUADPACK routine exposed through `scipy.integrate.quad`) in IEEE double precision, with the Riemann zeta function obtained from its analytic continuation as implemented in `scipy.special.zeta`. Starting from S_2, the partial sum was extended one term at a time and the integral recomputed after each increment.

The second quantity is the optimal error of Subsection 3.3, computed on the critical line as $d_n(1/2)^2 = \|E\|_{1/2}^2 - b^\top A^{-1} b$, with Gram data $A = (\langle G_j, G_k \rangle_{1/2})_{j,k=2}^n$ and $b = (\langle E, G_k \rangle_{1/2})_{k=2}^n$. Because E and the G_k have real Dirichlet coefficients,

$$\langle F, H \rangle_{1/2} = \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left[F(\tfrac{1}{2} + it)\, \overline{H(\tfrac{1}{2} + it)} \right] dt,$$

so A and b are real and the minimising coefficients may be taken real. These boundary integrals were evaluated on a fixed quadrature grid of approximately 2.8×10^4 nodes, refined near $t = 0$ and with a geometric tail out to $t = 3 \times 10^5$. On this grid the exact value $\|E\|_{1/2}^2 = 1/(2\alpha) = 1$ is reproduced to six digits, and the computed values of d_n^2 are unchanged, to plotting accuracy, under refinement of the grid; since the Gram matrix grows ill-conditioned with n (its condition number is the square of that of the underlying design matrix), the linear systems were solved only in the numerically stable range $n \le 5120$.

All computations were performed in Python 3.13 on a machine with a 5.2 GHz Intel Core i9-12900K processor and 128 GB of DDR4 RAM, running Ubuntu 22.04 LTS. The code generating all figures and tables of this section is available from the authors upon request.

4.2 Convergence cost across the strip

Our first experiment quantifies how the rate of convergence of (S_n) degrades as α approaches the critical line. We discretised the interval $[0.5, 1.0]$ into 100 equally spaced points,

$$\alpha_i = 0.5 + 0.005\,(i - 1), \qquad i = 1, \dots, 100,$$

and, for three tolerance levels $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, recorded for each α_i the smallest number of terms n for which $s_n < \varepsilon$, together with the time required to reach it. Here n serves as a machine-independent measure of how rapidly S_n approaches E: the larger the n needed to meet a fixed ε, the slower the convergence. To keep the computation feasible, each run was capped at a time budget of 3600 seconds, after which the run was terminated and the largest n attained recorded. The results are reported in Figure 1 and Table 1.

A sharp, tolerance-dependent transition is evident. The coarse target $\varepsilon = 10^{-2}$ is met for every $\alpha \ge 0.60$, requiring only six to twelve terms once $\alpha \ge 0.70$; the intermediate target $\varepsilon = 10^{-3}$ is met for $\alpha \ge 0.75$, and the fine target $\varepsilon = 10^{-4}$ only for $\alpha \ge 0.90$. The threshold in α below which the time budget is exhausted thus rises steadily as the demanded accuracy increases, and for $\alpha \le 0.55$ none of the three tolerances is reached within the allotted 3600 seconds.

Two features deserve emphasis. First, Theorem 3.1 guarantees convergence of S_n to E only for $\alpha > 1$, yet the experiments drive the error below 10^{-3} for α as small as ≈ 0.75: the analytic threshold is not sharp, and the data support the expectation that convergence persists well into the critical strip. Second, the failure to attain a prescribed tolerance at small α must be read with care, since it records a convergence rate that deteriorates rapidly as $\alpha \to 1/2$ and is equally consistent with divergence. Disentangling these two readings near the endpoints is the subject of the next subsection.

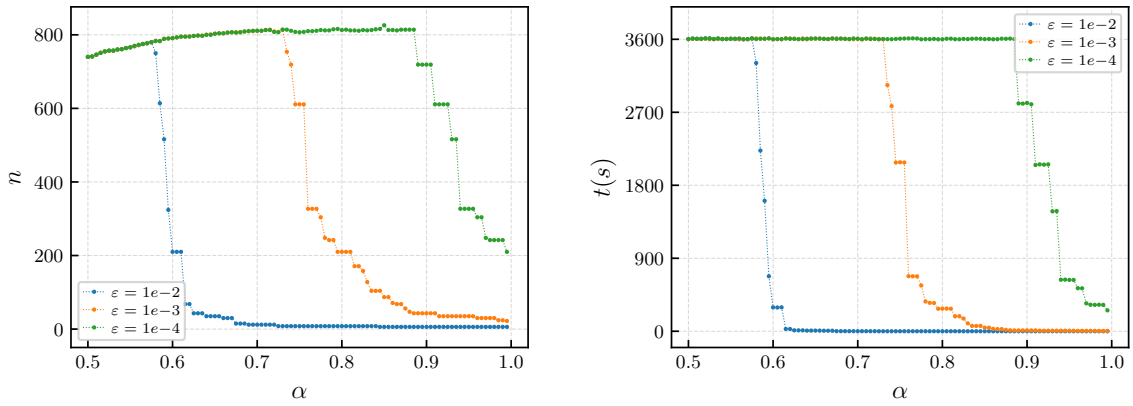


Figure 1: For each α, the smallest number of terms n (left) and the computation time (right) needed to reach $s_n < \varepsilon$, for $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$. Each run was capped at a 3600 s time budget.

α	$\varepsilon = 10^{-2}$		$\varepsilon = 10^{-3}$		$\varepsilon = 10^{-4}$	
	n	Time (s)	n	Time (s)	n	Time (s)
0.50	740	3.600×10^{3}	740	3.600×10^{3}	740	3.600×10^{3}
0.55	766	3.600×10^{3}	766	3.600×10^{3}	766	3.600×10^{3}
0.60	210	2.941×10^{2}	791	3.600×10^{3}	791	3.600×10^{3}
0.65	35	8.237	803	3.600×10^{3}	803	3.600×10^{3}
0.70	12	8.330×10^{-1}	811	3.600×10^{3}	811	3.600×10^{3}
0.75	8	3.520×10^{-1}	611	2.084×10^{3}	807	3.600×10^{3}
0.80	8	3.500×10^{-1}	210	2.792×10^{2}	813	3.600×10^{3}
0.85	6	1.760×10^{-1}	87	4.466×10^{1}	826	3.600×10^{3}
0.90	6	1.760×10^{-1}	43	1.057×10^{1}	719	2.815×10^{3}
0.95	6	1.760×10^{-1}	35	6.988	327	6.333×10^{2}

Table 1: Smallest number of terms n and time (in seconds) needed to reach $s_n < \varepsilon$, for selected α and $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$. Entries at the 3600 s cap indicate that the tolerance was not attained within the time budget. Cf. Figure 1.

4.3 Case studies at the endpoints $\alpha \approx 1/2$ and $\alpha \approx 1$

The transition identified in Subsection 4.2 separates two qualitatively distinct regimes. To examine each in isolation, we refine the analysis near the two endpoints of the interval. For $\alpha_{\min} \in \{0.5, 0.9\}$ we evaluate s_n on the grids

$$\alpha_i = \alpha_{\min} + 0.01\, i, \qquad i = 0, 1, \ldots, 9,$$

at the sample points $n = 1000\, j$, $j = 1, \ldots, 100$, so that the behaviour of s_n is tracked up to $n = 10^5$. This larger range of n lets the asymptotic trend of each sequence emerge, beyond the

onset captured by the tolerance experiment of the previous subsection. The resulting sequences are plotted in Figures 2 and 3, and tabulated in full in Appendix A (Tables 3 and 4).

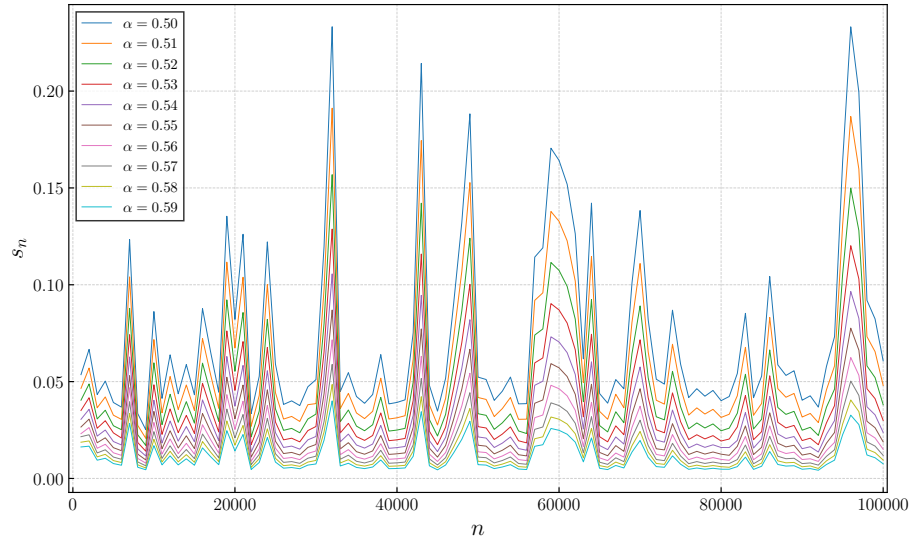


Figure 2: The approximation error s_n as a function of n, for $0.50 \leq \alpha \leq 0.59$ (divergent regime).

The divergent regime $0.50 \leq \alpha \leq 0.59$**.** Figure 2 and Table 3 show that, across this entire interval, s_n exhibits prominent oscillations with no stabilising trend: not only does the sequence fail to decay, but even its local minima remain bounded away from zero, hovering near 4×10^{-2} at $\alpha = 0.50$ while the peaks reach order 10^{-1}, recurring without attenuation up to $n = 10^5$. This is precisely the behaviour anticipated by theory, since the divergence of the $L^2(0,1)$ analogue of s_n proved by Báez-Duarte [3] transfers to $H^2(\mathbb{C}_{1/2})$ and forces $s_n \not\to 0$ at $\alpha = 1/2$. The numerics show that this obstruction is robust throughout the lower edge of the strip, and not confined to the critical line itself.

The convergent regime $0.90 \leq \alpha \leq 0.99$**.** Figure 3 and Table 4 reveal a markedly different picture. The sequence s_n now displays a clear overall decay, with oscillatory peaks whose amplitude diminishes steadily as n grows; although the decrease is not strictly monotone, the envelope of the sequence contracts consistently. For $\alpha = 0.99$, the error falls from 1.52×10^{-4} at $n = 10^3$ to 3.13×10^{-6} at $n = 10^5$, and the same qualitative behaviour holds uniformly across the interval. These observations strongly suggest that $S_n \to E$ in $H^2(\mathbb{C}_\alpha)$ throughout $0.90 \leq \alpha \leq 0.99$.

4.4 The optimal error on the critical line

The critical line is the one place in the range studied where S_n demonstrably fails: by Báez-Duarte [3] the Möbius coefficients do not sum to E at $\alpha = 1/2$, and the undamped oscillations of Figure 2 are the price of this fixed choice. Since it is exactly at $\alpha = 1/2$ that Theorem 2.2 formulates the RH, the natural question is what this failure rules out. The answer is: nothing

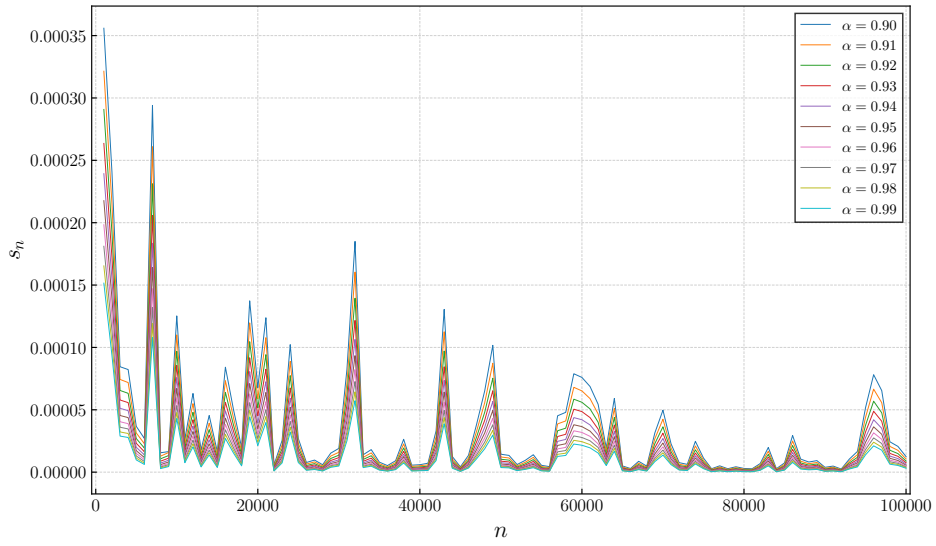


Figure 3: The approximation error s_n as a function of n, for $0.90 \leq \alpha \leq 0.99$ (convergent regime).

about the subspaces $\mathrm{span}\{G_2, \ldots, G_n\}$ themselves, since $\mu(k)$ is only one admissible choice of coefficients, and it is the subspaces, not the choice, that Theorem 3.2 ties to the zeros of ζ. This is what motivates the present experiment: we replace s_n by the optimal error $d_n(1/2)^2$ of Subsection 3.3, the true distance from E to $\mathrm{span}\{G_2, \ldots, G_n\}$ on the critical line, which removes the coefficients from the picture by optimising over all of them. It is computed by the Gram matrix method of Subsection 4.1.

Figure 4 contrasts the two sequences, and Table 2 records their values at selected n. While the Möbius error s_n remains near $10^{-1.5}$ with persistent oscillations, the optimal error d_n^2 decreases steadily, though very slowly, and satisfies $d_n^2 \leq s_n$ throughout the computed range. The "divergence" on the critical line is therefore a statement about the coefficients rather than about the subspaces: the obstruction lies in the fixed choice $c_k = \mu(k)$, while the true distance from E to $\mathrm{span}\{G_2, \ldots, G_n\}$ continues to decrease over the range accessible to stable computation.

n	$d_n(1/2)^2$	$s_n(1/2)$
40	1.273×10^{-2}	1.442×10^{-1}
160	9.355×10^{-3}	6.846×10^{-2}
640	7.022×10^{-3}	3.657×10^{-2}
1280	6.361×10^{-3}	3.473×10^{-2}
2560	5.931×10^{-3}	4.312×10^{-2}
5120	5.282×10^{-3}	3.850×10^{-2}

Table 2: The optimal error $d_n(1/2)^2$ and the Möbius error $s_n(1/2)$ at selected n. In every case $d_n^2 \leq s_n$.

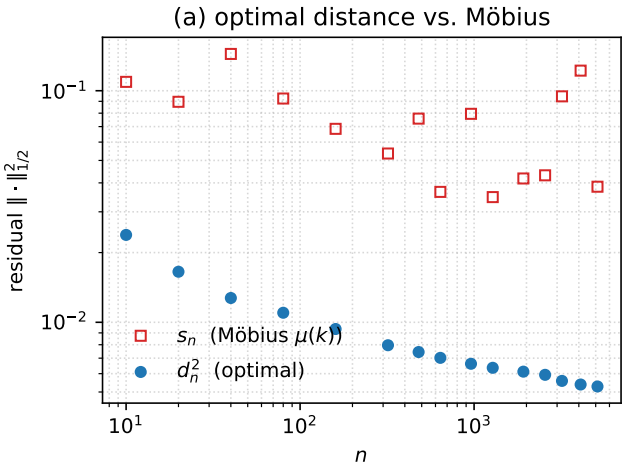


Figure 4: The optimal approximation error $d_n(1/2)^2$ (blue) versus the Möbius error s_n (red) on the critical line $\alpha = 1/2$, for $2 \le n \le 5120$, displayed in log–log scale.

4.5 Concluding discussion

The three experiments paint a consistent picture. Across the strip, the cost of approximating E by the partial sums S_n increases sharply as α decreases towards $1/2$, with a tolerance-dependent transition (Subsection 4.2). Near the upper edge, the error s_n decays steadily over the whole computed range (Subsection 4.3), supporting the expectation that the convergence established in Theorem 3.1 for $\alpha > 1$ persists well into the critical strip. On the critical line, where S_n fails, the obstruction is attributable to the Möbius coefficients rather than to the subspaces themselves, since the optimal error $d_n(1/2)^2$ – the quantity that Theorem 3.2 ties to a zero of ζ in $\mathbb{C}_\alpha$ – continues to decrease (Subsection 4.4). Together, these observations exhibit the interplay between approximation quality and the zero distribution of ζ.

We stress, finally, the conditional nature of this reading. By Proposition 3.1, genuine convergence $S_n \to E$ for some $1/2 \le \alpha < 1$ would place E in the closed span of $\{G_k : k \ge 2\}$ and hence furnish a zero-free half-plane $\mathbb{C}_\alpha$ deep within the critical strip, a conclusion lying far beyond all currently known zero-free regions and widely regarded as being as difficult as the Riemann Hypothesis itself. The evidence presented here is therefore numerical and suggestive, not a proof.

A Tabulated data

This appendix records the values of the Möbius error s_n underlying the case studies of Subsection 4.3: Table 3 for the divergent regime $0.50 \le \alpha \le 0.59$ (cf. Figure 2), and Table 4 for the convergent regime $0.90 \le \alpha \le 0.99$ (cf. Figure 3).

n	$\alpha = 0.50$	$\alpha = 0.51$	$\alpha = 0.52$	$\alpha = 0.53$	$\alpha = 0.54$
1000	5.358×10^{-2}	4.651×10^{-2}	4.039×10^{-2}	3.510×10^{-2}	3.050×10^{-2}
10000	8.618×10^{-2}	7.170×10^{-2}	5.968×10^{-2}	4.836×10^{-2}	4.037×10^{-2}
20000	8.215×10^{-2}	6.742×10^{-2}	5.537×10^{-2}	4.549×10^{-2}	3.739×10^{-2}
30000	5.103×10^{-2}	3.865×10^{-2}	3.322×10^{-2}	2.682×10^{-2}	2.169×10^{-2}
40000	3.926×10^{-2}	3.131×10^{-2}	2.498×10^{-2}	1.994×10^{-2}	1.592×10^{-2}
50000	5.233×10^{-2}	4.180×10^{-2}	3.343×10^{-2}	2.674×10^{-2}	2.143×10^{-2}
60000	1.642×10^{-1}	1.328×10^{-1}	1.075×10^{-1}	8.699×10^{-2}	7.050×10^{-2}
70000	1.384×10^{-1}	1.110×10^{-1}	8.912×10^{-2}	7.162×10^{-2}	5.758×10^{-2}
80000	4.016×10^{-2}	3.152×10^{-2}	2.461×10^{-2}	1.933×10^{-2}	1.598×10^{-2}
90000	4.038×10^{-2}	3.171×10^{-2}	2.490×10^{-2}	1.955×10^{-2}	1.537×10^{-2}
100000	6.073×10^{-2}	4.805×10^{-2}	3.805×10^{-2}	2.996×10^{-2}	2.377×10^{-2}
n	$\alpha = 0.55$	$\alpha = 0.56$	$\alpha = 0.57$	$\alpha = 0.58$	$\alpha = 0.59$
1000	2.655×10^{-2}	2.313×10^{-2}	2.155×10^{-2}	1.870×10^{-2}	1.625×10^{-2}
10000	3.461×10^{-2}	2.891×10^{-2}	2.417×10^{-2}	2.022×10^{-2}	1.692×10^{-2}
20000	3.077×10^{-2}	2.534×10^{-2}	2.089×10^{-2}	1.723×10^{-2}	1.421×10^{-2}
30000	1.753×10^{-2}	1.419×10^{-2}	1.149×10^{-2}	9.308×10^{-3}	7.468×10^{-3}
40000	1.272×10^{-2}	1.016×10^{-2}	8.060×10^{-3}	6.438×10^{-3}	5.149×10^{-3}
50000	1.716×10^{-2}	1.376×10^{-2}	1.103×10^{-2}	8.851×10^{-3}	7.112×10^{-3}
60000	5.715×10^{-2}	4.633×10^{-2}	3.762×10^{-2}	3.057×10^{-2}	2.486×10^{-2}
70000	4.633×10^{-2}	3.731×10^{-2}	3.009×10^{-2}	2.428×10^{-2}	1.960×10^{-2}
80000	1.253×10^{-2}	9.831×10^{-3}	7.715×10^{-3}	6.061×10^{-3}	4.762×10^{-3}
90000	1.208×10^{-2}	9.757×10^{-3}	7.671×10^{-3}	6.034×10^{-3}	4.750×10^{-3}
100000	1.887×10^{-2}	1.508×10^{-2}	1.191×10^{-2}	9.475×10^{-3}	7.543×10^{-3}

Table 3: Values of the Möbius error s_n at the sampled n, for $0.50 \le \alpha \le 0.59$ (divergent regime). Cf. Figure 2.

n	$\alpha = 0.90$	$\alpha = 0.91$	$\alpha = 0.92$	$\alpha = 0.93$	$\alpha = 0.94$
1000	3.558×10^{-4}	3.215×10^{-4}	2.909×10^{-4}	2.636×10^{-4}	2.392×10^{-4}
10000	1.253×10^{-4}	1.100×10^{-4}	9.691×10^{-5}	8.562×10^{-5}	7.583×10^{-5}
20000	6.772×10^{-5}	5.882×10^{-5}	5.149×10^{-5}	4.498×10^{-5}	3.939×10^{-5}
30000	1.905×10^{-5}	1.594×10^{-5}	1.360×10^{-5}	1.164×10^{-5}	9.845×10^{-6}
40000	5.966×10^{-6}	5.138×10^{-6}	4.214×10^{-6}	3.460×10^{-6}	2.849×10^{-6}
50000	1.437×10^{-5}	1.215×10^{-5}	1.032×10^{-5}	8.802×10^{-6}	7.528×10^{-6}
60000	7.597×10^{-5}	6.516×10^{-5}	5.604×10^{-5}	4.854×10^{-5}	4.202×10^{-5}
70000	4.970×10^{-5}	4.245×10^{-5}	3.640×10^{-5}	3.125×10^{-5}	2.697×10^{-5}
80000	2.970×10^{-6}	2.375×10^{-6}	1.891×10^{-6}	1.509×10^{-6}	1.208×10^{-6}
90000	3.872×10^{-6}	3.146×10^{-6}	2.564×10^{-6}	2.095×10^{-6}	1.720×10^{-6}
100000	1.248×10^{-5}	1.051×10^{-5}	8.904×10^{-6}	7.568×10^{-6}	6.457×10^{-6}
n	$\alpha = 0.95$	$\alpha = 0.96$	$\alpha = 0.97$	$\alpha = 0.98$	$\alpha = 0.99$
1000	2.177×10^{-4}	1.983×10^{-4}	1.812×10^{-4}	1.654×10^{-4}	1.516×10^{-4}
10000	6.707×10^{-5}	5.974×10^{-5}	5.335×10^{-5}	4.775×10^{-5}	4.297×10^{-5}
20000	3.463×10^{-5}	3.051×10^{-5}	2.699×10^{-5}	2.395×10^{-5}	2.133×10^{-5}
30000	8.498×10^{-6}	7.364×10^{-6}	6.408×10^{-6}	5.598×10^{-6}	4.909×10^{-6}
40000	2.350×10^{-6}	1.945×10^{-6}	1.616×10^{-6}	1.347×10^{-6}	1.127×10^{-6}
50000	6.473×10^{-6}	5.585×10^{-6}	4.839×10^{-6}	4.212×10^{-6}	3.683×10^{-6}
60000	3.650×10^{-5}	3.184×10^{-5}	2.781×10^{-5}	2.446×10^{-5}	2.157×10^{-5}
70000	2.335×10^{-5}	2.031×10^{-5}	1.772×10^{-5}	1.553×10^{-5}	1.367×10^{-5}
80000	9.647×10^{-7}	7.736×10^{-7}	6.219×10^{-7}	5.007×10^{-7}	4.044×10^{-7}
90000	1.416×10^{-6}	1.171×10^{-6}	9.733×10^{-7}	8.124×10^{-7}	6.820×10^{-7}
100000	5.534×10^{-6}	4.763×10^{-6}	4.117×10^{-6}	3.576×10^{-6}	3.125×10^{-6}

Table 4: Values of the Möbius error s_n at the sampled n, for $0.90 \leq \alpha \leq 0.99$ (convergent regime). Cf. Figure 3.

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